

RTKNAVI-BH User Manual

RTKNAVI-BH: An Open-Source Extension of RTKNAVI to Enable Real-Time PPP with PPP-B2b and HAS Corrections

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1 Introduction

PPP-B2b and HAS are two freely available satellite-based augmentation services that enable real-time precise point positioning (PPP) without relying on ground communication networks. PPP-B2b, provided by the BeiDou Navigation Satellite System-3 (BDS-3), broadcasts real-time orbit and clock corrections for GPS and BDS satellites via the B2b signal, covering China and its surrounding areas (CSNO, 2020, 2021). HAS, provided by the European Galileo, delivers real-time orbit, clock, and code bias corrections for GPS and Galileo satellites via the E6-B signal with global coverage (European Union, 2022, 2023). Both services have been demonstrated to achieve decimeter-level kinematic positioning accuracy (Nie et al., 2021; Xu et al., 2021; Zhang et al., 2022; Liu et al., 2022; Naciri et al., 2023; Zhou et al., 2023), making them viable new approaches for real-time PPP applications.

Given the channel bandwidth limitations of satellite links, both PPP-B2b and HAS have abandoned the standard RTCM SSR format and instead defined their own dedicated frame structures optimized for transmission efficiency. This incompatibility prevents mainstream open-source GNSS software from directly decoding and using these corrections. To facilitate the evaluation and practical use of these satellite-based augmentation services, several Python-based decoders have been developed, such as GHASP (Gioia et al., 2022), HASlib (Horst, 2021), and NavDecoder (Zhao, 2025). Although these tools provide decoding capabilities, they are all implemented in Python. Since most prominent open-source GNSS software is developed in C/C++, these Python-based decoders cannot be directly embedded into existing applications and must operate as external processes when performing PPP with PPP-B2b and HAS corrections. This not only introduces operational inconvenience but also causes data transfer latency. To address this integration issue, Liu et al. (2026) developed RTKLIB-B2b, an open-source C/C++ toolkit built upon RTKLIB. It supports the real-time decoding of PPP-B2b corrections and navigation messages from SinoGNSS and Unicore receivers, further enabling real-time PPP. Validation experiments confirmed that RTKLIB-B2b can successfully provide decimeter-level kinematic PPP. However, RTKLIB-B2b is limited to the PPP-B2b service and relies on command-line configuration without a graphical user interface (GUI).

To address these limitations and advance the research and application of PPP-B2b and HAS, an open-source C/C++ extension of RTKNAVI, named RTKNAVI-BH, has been developed. While preserving all native functionalities and the GUI of the base software, the extended program incorporates a dedicated SinoGNSS receiver module capable of parsing the proprietary serial protocol

and extracting observations, broadcast ephemerides, and PPP-B2b/HAS correction messages. The decoded corrections are directly injected into the internal data structures of the existing PPP engine. This facilitates both real-time and simulated real-time positioning within a single executable, thereby eliminating the integration complexity and potential latency drawbacks associated with external decoding solutions. Given that RTKLIB is among the most widely used open-source software for GNSS data processing, building our extensions on this platform substantially lowers the barrier to learning, research, and practical deployment, while ensuring seamless compatibility with the existing RTKLIB ecosystem.

The remainder of this manual is organized as follows. Chapter 2 describes the methodology for recovering precise orbit and clock offsets with PPP-B2b and HAS corrections. Chapter 3 introduces the project file structure and source files of RTKNAVI-BH. Chapter 4 covers the installation, compilation, execution, and packaging of RTKNAVI-BH. Chapter 5 presents the learning project in detail, including configuration, output formats, and sample real-time PPP results.

2 Satellite precise orbit and clock offset recovery with PPP-B2b and HAS corrections

2.1 Recovery with PPP-B2b corrections

In the PPP-B2b service, satellite position corrections are applied by adjusting the broadcast ephemeris position using an orbit error correction vector (CSNO, 2020). Moreover, the corrections refer to the LNAV navigation messages for GPS and the CNAV1 navigation messages for BDS-3, respectively. If the issue of Data of Navigation (IODN) associated with the orbit error correction matches the Issue of Data of the Clock (IODC) from the broadcast ephemeris, the precise satellite orbit X_{B2b} can be determined as follows:

$$X_{B2b} = X_{brdc} - \Delta X_{cor} \quad (1)$$

where X_{brdc} refers to the satellite position obtained from the broadcast ephemeris, and ΔX_{cor} represents the satellite orbit correction vector. The correction vector ΔX_{cor} is computed according to the following formula:

$$\begin{aligned} \Delta X_{cor} &= [e_R \quad e_A \quad e_C] \cdot \Delta O_{B2b} \\ e_R &= \frac{r}{|r|} \\ e_A &= \frac{r}{|r|} \\ e_C &= \frac{r \times \dot{r}}{|r \times \dot{r}|} \end{aligned} \quad (2)$$

$$e_A = e_C \times e_R$$

where ΔO_{B2b} represents the PPP-B2b orbit corrections expressed in radial (R), along-track (A), and cross-track (C) directions; $\dot{r}$ and r denote the satellite velocity and position vectors, respectively, which are computed based on the broadcast ephemeris.

The PPP-B2b clock offset correction ΔC_{B2b} is used to adjust the broadcast clock offset t_{brdc} , obtaining the precise satellite clock offset t_{B2b} :

$$t_{B2b} = t_{brdc} - \frac{\Delta C_{B2b}}{c} \quad (3)$$

where c is the velocity of light in vacuum.

To eliminate inter-frequency code biases, PPP-B2b also provides differential code bias (DCB) corrections, with only BDS-3 DCB corrections currently available. The corrected pseudorange $\tilde{P}_{sig}$ on the sig channel can be obtained as:

$$\tilde{P}_{sig} = P_{sig} - DCB_{sig} \quad (4)$$

where P_{sig} is the raw pseudorange captured by the receiver.

2.2 Recovery with HAS corrections

Galileo HAS provides satellite orbit, clock offset corrections and code biases. The corrections refer to the LNAV navigation messages for GPS and I/NAV navigation messages for Galileo, respectively (European Union, 2022).

The precise satellite orbits X_{HAS} recovered with HAS corrections ΔX_{cor} can be formulated as:

$$X_{HAS} = X_{brdc} + \Delta X_{cor} \quad (5)$$

where ΔX_{cor} can be obtained according to the following formula:

$$\begin{aligned} \Delta X_{cor} &= [e_R \quad e_A \quad e_C] \cdot \Delta O_{HAS} \\ e_A &= \frac{\dot{r}}{|\dot{r}|} \\ e_C &= \frac{r \times \dot{r}}{|r \times \dot{r}|} \\ e_R &= e_A \times e_C \end{aligned} \quad (6)$$

where ΔO_{HAS} is the HAS orbit correction vector expressed in R, A and C components.

The precise satellite clock offset t_{HAS} recovered with the correction ΔC_{HAS} can be obtained by:

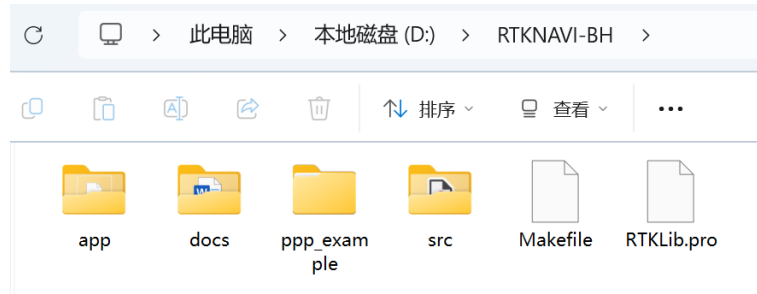
$$t_{HAS} = t_{brdc} + \frac{\Delta C_{HAS}}{c} \quad (7)$$

And the pseudorange can be corrected as follows:

$$\tilde{P}_{sig} = P_{sig} + DCB_{sig} \quad (8)$$

3 Download

No additional libraries need to be downloaded. The source code, executable files, and dependencies of RTKNAVI-BH are all contained within the “RTKNAVI-BH” folder. The main contents of this folder are as follows:



The “RTKNAVI-BH\src” folder stores the source code for processing raw binary data from SinoGNSS receivers, including raw observations, broadcast ephemerides, and PPP-B2b/HAS correction messages. The key files in this folder are listed below:

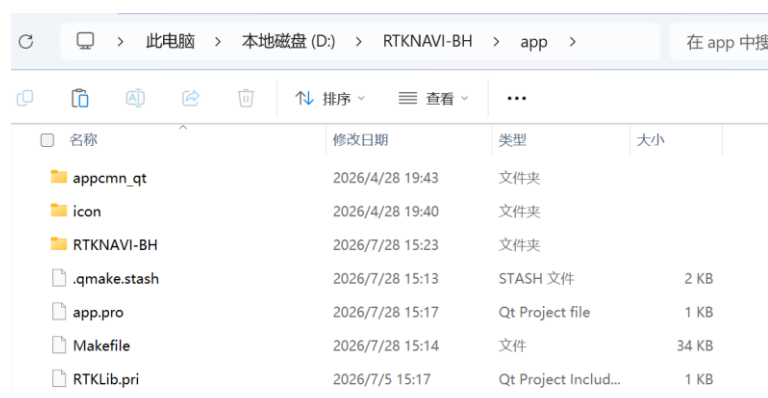
名称	修改日期	类型	大小
pntpos.c	2026/7/16 11:30	C Source	25 KB
ephemeris.c	2026/7/16 11:29	C Source	39 KB
has.c	2026/7/16 11:17	C Source	396 KB
sino.c	2026/7/16 10:41	C Source	36 KB
b2b.c	2026/7/16 10:40	C Source	20 KB
rtkpos.c	2026/7/2 20:42	C Source	68 KB
rtklib.h	2026/7/2 20:42	C/C++ Header	98 KB
options.c	2026/4/27 10:52	C Source	22 KB
src.pro	2026/4/24 11:09	Qt Project file	2 KB
iersmodel.c	2026/1/20 15:33	C Source	28 KB
tides.c	2026/1/17 8:50	C Source	11 KB

- sino.c: SinoGNSS receiver data stream processing module. It parses the proprietary serial protocol, identifies message types, and dispatches observations and broadcast ephemerides to the standard RTKLIB binary decoder, while forwarding PPP-B2b and HAS messages to their dedicated decoding modules.
- b2b.c: PPP-B2b correction processing module. It decodes the PPP-B2b messages and stores

the recovered corrections in the newly added b2bsat member of the nav structure.

- has.c: HAS correction processing module. It accumulates and HPVRS-decodes the HAS messages and stores the recovered corrections in the newly added hassat member of the nav structure.
- Other files originate from the standard RTKLIB distribution. Minor modifications have been made to accommodate the newly added features without breaking original functionalities. All newly added and modified code is marked with the [RTKNAVI-BH] comment tag for easy identification.

The “RTKNAVI-BH\app” directory contains all source files required to build the RTKNAVI-BH graphical program. It consists of three subfolders:



- appcmn_qt: Stores shared Qt GUI common source code for RTKLIB graphical applications. This module implements universal functions including dialog windows, file selection utilities, format conversion between RTKLIB C-style data structures and Qt components, and configuration read/write interfaces.
- icon: Holds icon image resources used by the graphical interface of RTKNAVI-BH.
- RTKNAVI-BH: Contains the main program source code of the software, which implements the core graphical interface logic, real-time data visualization, parameter configuration panel, and positioning result display functions.

The “RTKNAVI-BH\docs” folder contains this user manual.

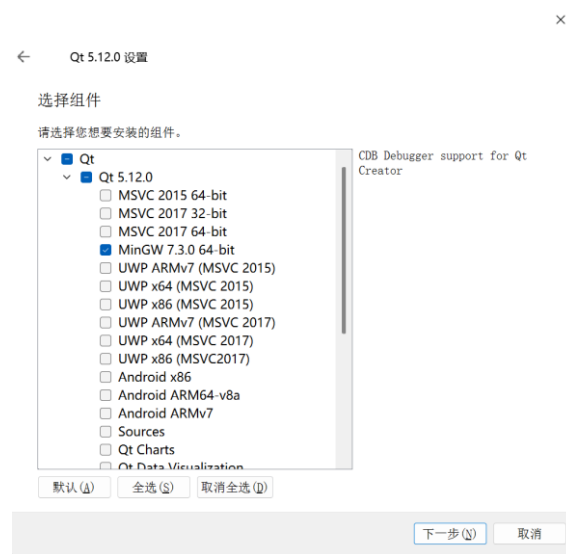
The “RTKNAVI-BH\ppp_example” folder stores raw data collected from SinoGNSS receivers and corresponding positioning results. The raw data include observations, broadcast ephemerides, PPP-B2b and HAS correction messages, which can be directly used for PPP positioning test with PPP-B2b or HAS services in this software.

4 Installation

4.1 Install Qt

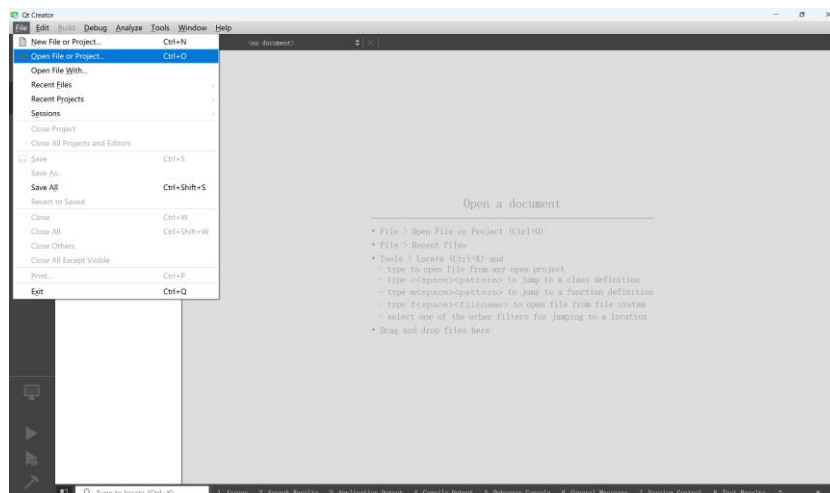
RTKNAVI-BH was developed in the C/C++ using the cross-platform Qt framework. It can be compiled and executed on popular operating systems such as Windows and Linux. We recommend debugging the software in Qt Creator.

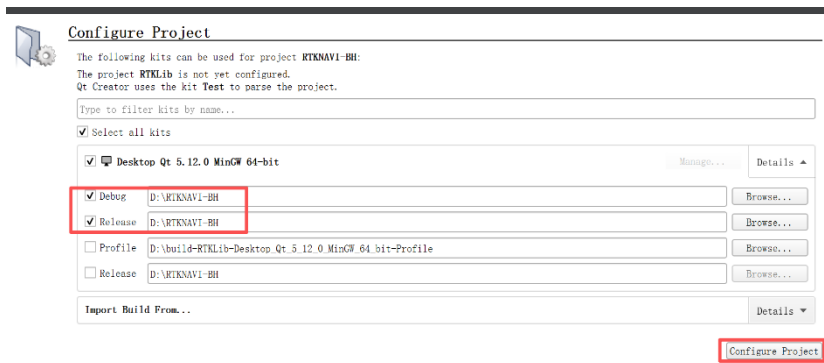
Users need to install Qt5 (version 5.12.0 or later) from the official Qt website (www.qt.io) to open and build the project. During the installation of Qt5.12.0, select **MinGW 7.3.0 64-bit** as the compiler, matching your operating system.



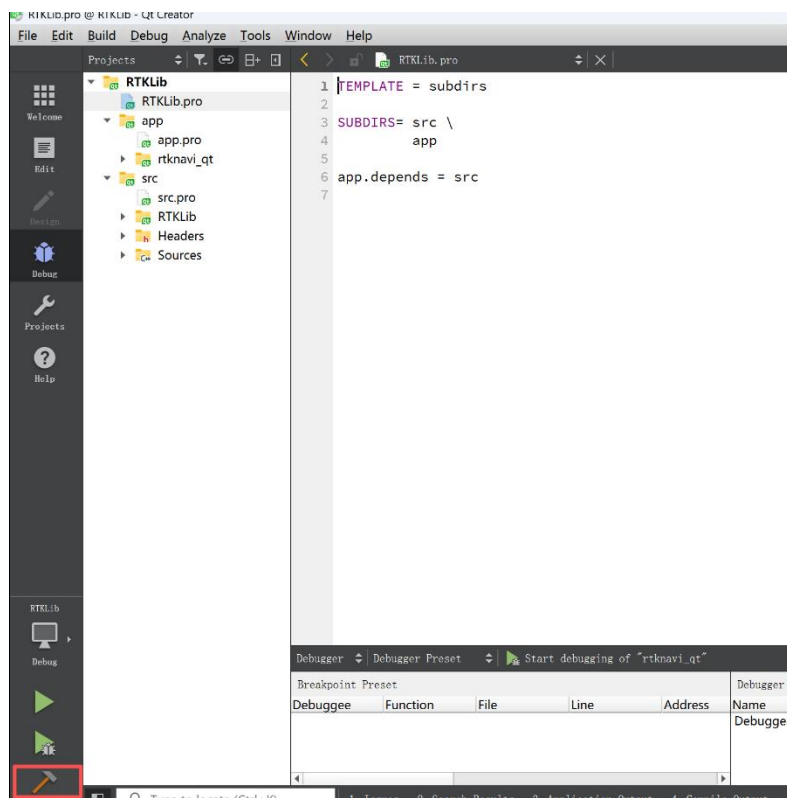
4.2 Build and run

After the Qt installation is complete, launch Qt Creator. Select “File > Open File or Project...” from the menu bar to open the “RTKLib.pro” project file in the “RTKNAVI-BH” folder. Follow the instructions below to configure the project:

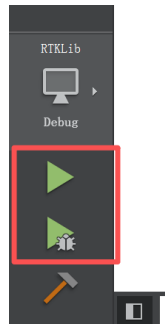




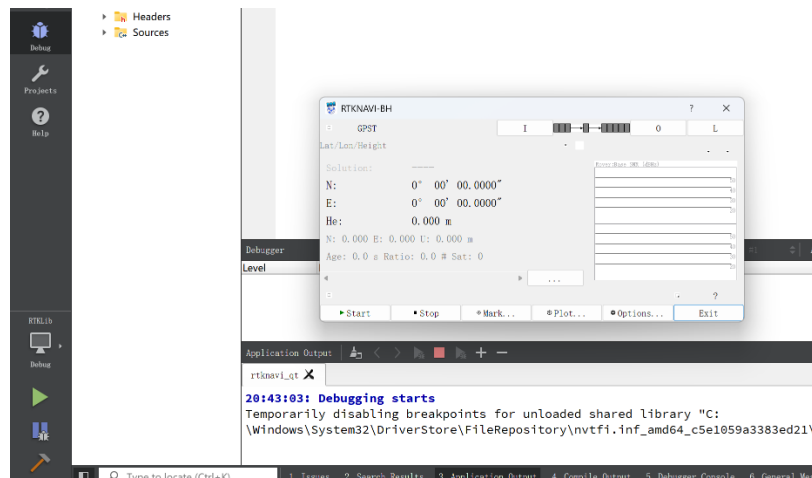
Once the project is configured, you can start the build process by clicking the build button (indicated by a red circle in the image below). The default build mode is Debug. You can switch to Release mode by clicking the monitor icon above the build button. Note that a full rebuild is required after switching modes. The Release mode generates optimized binaries ready for packaging.



To run the program, click the first triangular Run button to execute without debugging, or the second triangular button to start debugging.



After clicking the Run button, the main window of RTKNAVI-BH will appear, as shown in the figure below.



4.3 Package executable program

You can create a standalone executable package by following the steps below.

1. Generate the Release executable:

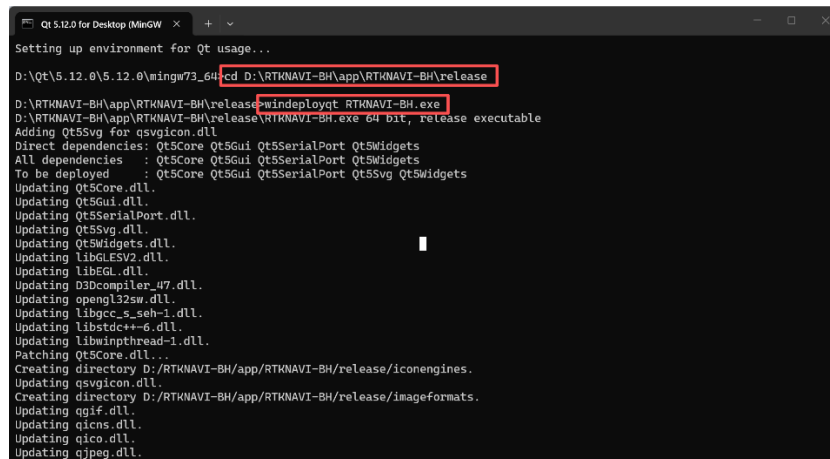
Ensure the project is built successfully in Release mode in Qt Creator. This generates the executable file.



2. Package program

For Windows users: The executable “RTKNAVI-BH.exe” is generally located in the path

“\RTKNAVI-BH\app\RTKNAVI-BH\release”. Open the Qt command-line tool, navigate to this directory using the cd command, and run the deployment command “windeployqt RTKNAVI-BH.exe”. This tool automatically copies all required runtime dependencies into the release folder. You can then migrate this entire folder as a complete unit and run it on any Windows device without requiring a pre-installed Qt environment.



```

Qt 5.12.0 for Desktop (MinGW)
Setting up environment for Qt usage...
D:\Qt\5.12.0\5.12.0\mingw73_64>cd D:\RTKNAVI-BH\app\RTKNAVI-BH\release
D:\RTKNAVI-BH\app\RTKNAVI-BH\release>windeployqt RTKNAVI-BH.exe
D:\RTKNAVI-BH\app\RTKNAVI-BH\release>
Adding Qt5Svg for qsvgicon.dll
Direct dependencies: Qt5Core Qt5Gui Qt5SerialPort Qt5Widgets
All dependencies   : Qt5Core Qt5Gui Qt5SerialPort Qt5Widgets
To be deployed     : Qt5Core Qt5Gui Qt5SerialPort Qt5Svg Qt5Widgets
Updating Qt5Core.dll.
Updating Qt5Gui.dll.
Updating Qt5SerialPort.dll.
Updating Qt5Svg.dll.
Updating Qt5Widgets.dll.
Updating libGLESv2.dll.
Updating libEGL.dll.
Updating D3Dcompiler_47.dll.
Updating opengl32sw.dll.
Updating libgcc_s_seh-1.dll.
Updating libstdc++-6.dll.
Updating libwinpthread-1.dll.
Patching Qt5Core.dll...
Creating directory D:\RTKNAVI-BH\app\RTKNAVI-BH\release\iconengines.
Updating qsvgicon.dll.
Creating directory D:\RTKNAVI-BH\app\RTKNAVI-BH\release\imageformats.
Updating qgif.dll.
Updating qimageformats.dll.
Updating qico.dll.
Updating qjpeg.dll.

```

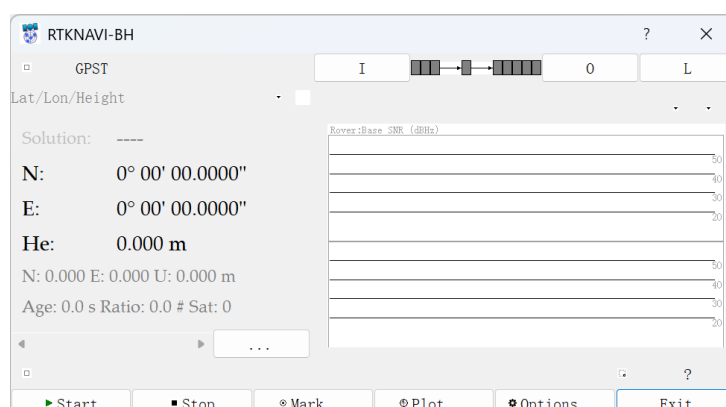
For Linux users: You can retrieve the full list of runtime dependencies for the executable using the “ldd” command, then manually package these dependent libraries together with the executable file to enable cross-device migration and direct execution.

5 Configuration example and results

This section provides a complete learning project that demonstrates how to perform real-time PPP positioning using different correction strategies: PPP-B2b alone, HAS alone, and a combination of both services. Follow the steps below to configure the software and start real-time processing.

5.1 Launching the software

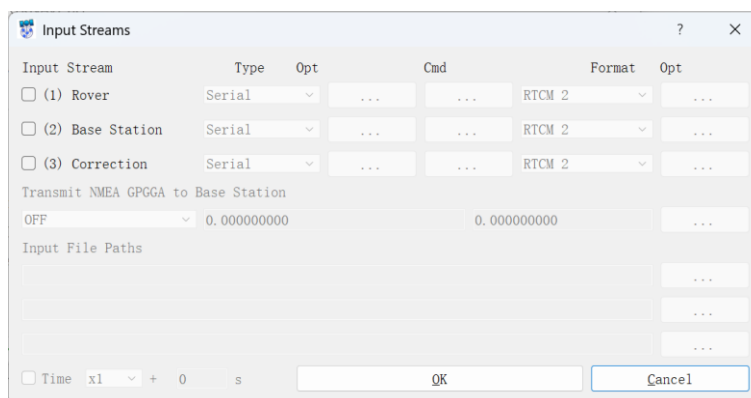
Compile and run RTKNAVI-BH as described in Section 4. The main window of the software is shown as follows:



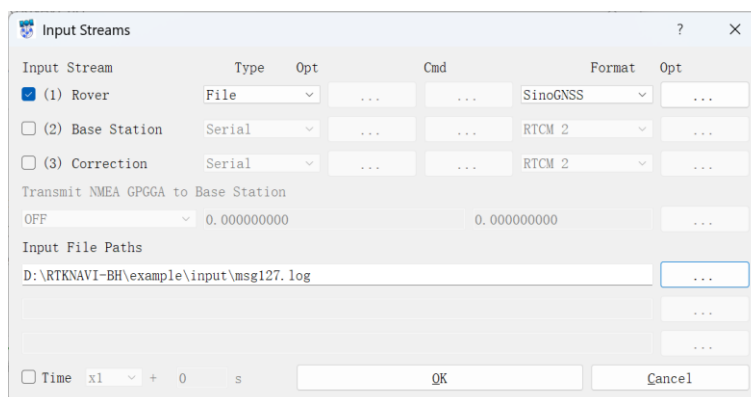
5.2 Configuring the input stream

Click the “I” button on the main window to open the input configuration dialog. For PPP, only the Rover input needs to be configured:

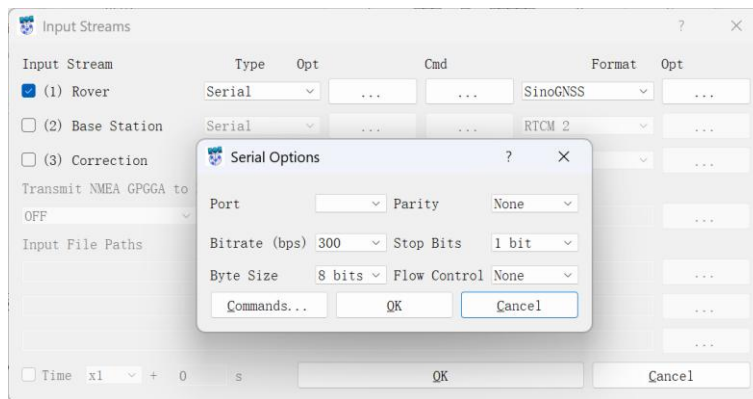
- **Type:** Select Serial if data is being collected by a receiver in real time. If using a pre-recorded data file, select File and specify the file path.
- **Format:** Choose SinoGNSS to enable parsing of the SinoGNSS proprietary protocol.
- **Options and Commands:** After selecting Serial, click the option button next to it to configure the serial port options and receiver commands. Enter the appropriate commands in the Commands field for the required data types. An example command set for collecting observations, broadcast ephemerides, and PPP-B2b/HAS corrections with a SinoGNSS receiver is shown in the figure. These commands instruct the receiver to output HAS corrections, PPP-B2b corrections, broadcast ephemerides for GPS, Galileo, and BDS, and raw observations. For commands related to other message types, refer to the SinoGNSS board reference manual.



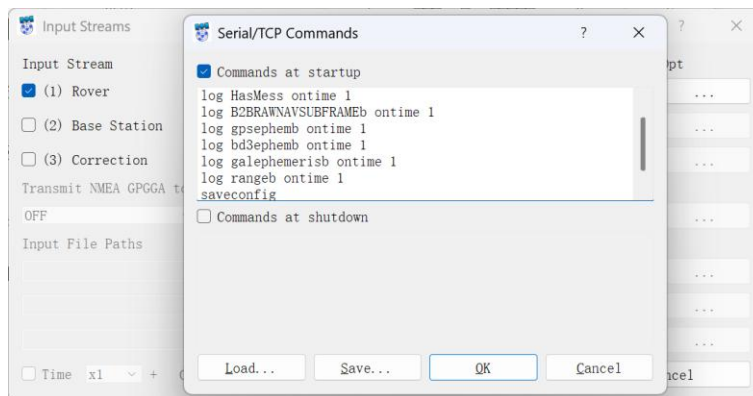
Input stream configuration dialog for PPP:



Serial port options configuration:



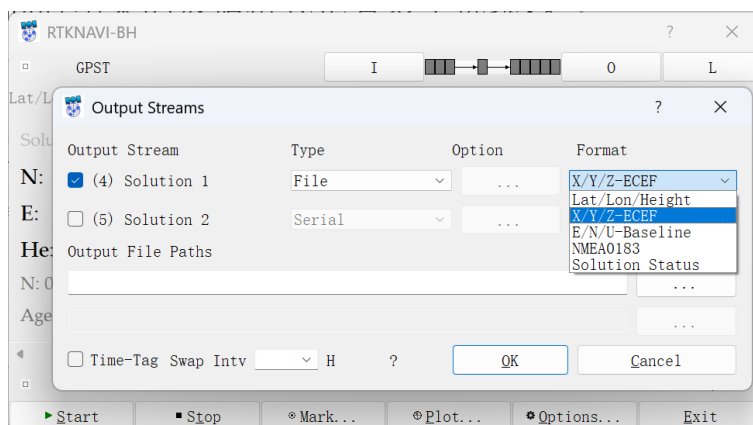
Example receiver commands for data collection:



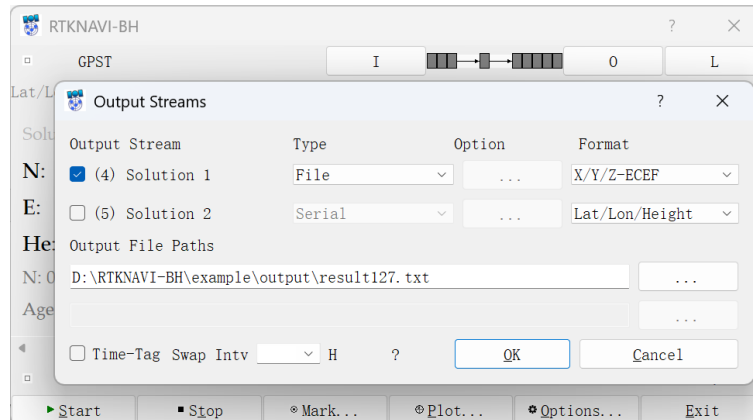
5.3 Configuring the output stream

Click the “O” button on the main window to open the output configuration dialog. Two output paths can be configured, corresponding to the Rover and Base solutions. For this project, only the first Solution output path needs to be set:

- **Type:** Typically select File to save the positioning results to a local file.
- **Format:** Select the desired output format from the drop-down list; several data formats are supported.

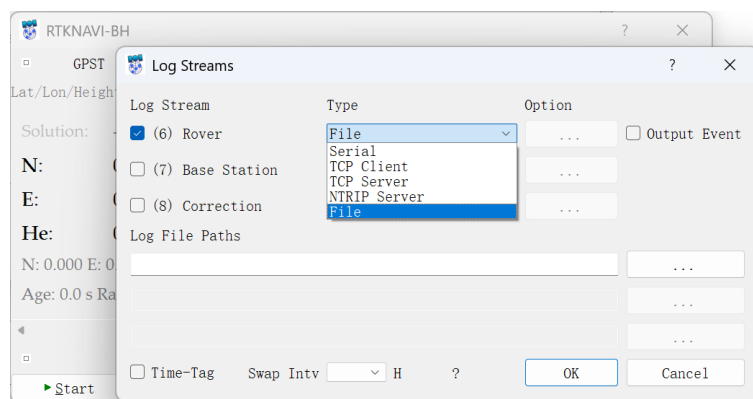


Output stream configuration dialog for PPP:



5.4 Configuring data logging (optional)

Click the “L” button on the main window to open the logging configuration dialog. This step enables saving the incoming raw data to a file. It is recommended to enable logging when performing real-time processing, so that the data can be replayed for post-processing and analysis later.



5.5 Configuring positioning parameters

After completing the input and output configurations, click the “Options” button at the bottom right of the main window to open the positioning parameter dialog. The main parameters for this project are set as follows:

- **Positioning Mode:** PPP Kinematic
- **Elevation Mask:** 10°
- **Earth Tides Correction:** Solid tide + Ocean loading + Pole tide
- **Ionosphere Correction:** Ionosphere-Free LC
- **Troposphere Correction:** Estimate ZTD
- **Satellite Ephemeris:** Select the correction source via the drop-down menu. Choose **Broadcast + B2b** for PPP-B2b alone, **Broadcast + HAS** for HAS alone, or **Broadcast + BHC** for the

combined strategy. In BHC mode, GPS and BDS satellites use PPP-B2b corrections, while Galileo satellites use HAS corrections. When switching between different correction strategies, the selected satellite constellations must be adjusted accordingly. The configuration interfaces for the three strategies are shown in the figures.

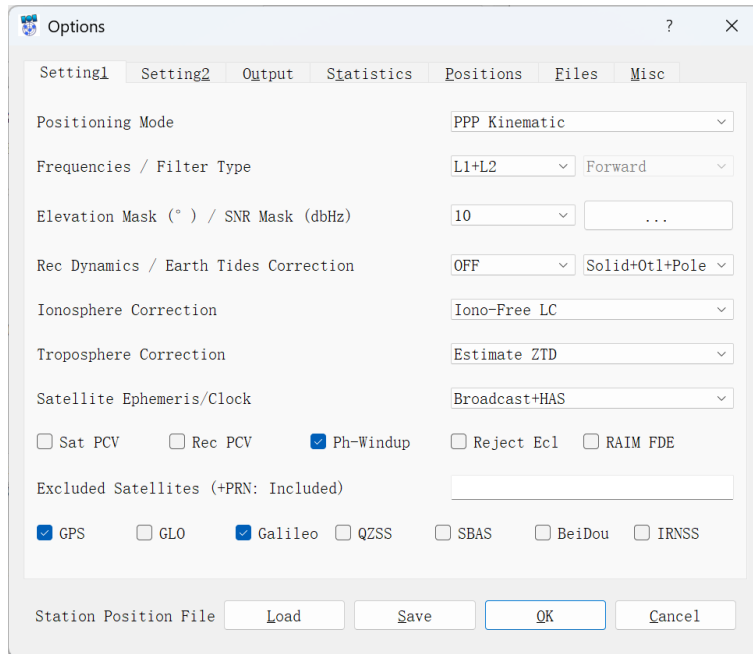
Other options are kept at their default values. Users may adjust them according to their specific application scenarios.

The screenshot shows the 'Options' dialog box with the 'Setting1' tab selected. The 'Positioning Mode' is set to 'Single'. 'Frequencies / Filter Type' is 'L1+L2' and 'Forward'. 'Elevation Mask (°) / SNR Mask (dbHz)' is '0'. 'Rec Dynamics / Earth Tides Correction' is 'OFF'. 'Ionosphere Correction' is 'Broadcast'. 'Troposphere Correction' is 'Saastamoinen'. 'Satellite Ephemeris/Clock' is 'Broadcast'. Under 'Excluded Satellites (+PRN: Included)', 'GPS' is checked, while 'GLO', 'Galileo', 'QZSS', 'SBAS', 'BeiDou', and 'IRNSS' are unchecked. The 'Station Position File' section has 'Load', 'Save', 'OK', and 'Cancel' buttons.

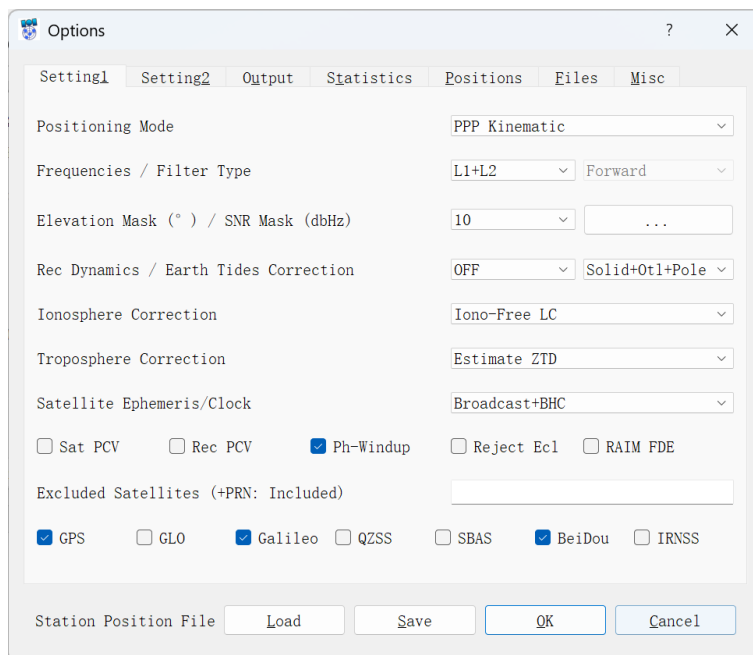
Satellite and ephemeris settings for PPP-B2b alone strategy:

The screenshot shows the 'Options' dialog box with the 'Setting1' tab selected. The 'Positioning Mode' is set to 'PPP Kinematic'. 'Frequencies / Filter Type' is 'L1+L2' and 'Forward'. 'Elevation Mask (°) / SNR Mask (dbHz)' is '10'. 'Rec Dynamics / Earth Tides Correction' is 'OFF'. 'Ionosphere Correction' is 'Iono-Free LC'. 'Troposphere Correction' is 'Estimate ZTD'. 'Satellite Ephemeris/Clock' is 'Broadcast+B2b'. Under 'Excluded Satellites (+PRN: Included)', 'GPS' and 'BeiDou' are checked, while 'GLO', 'Galileo', 'QZSS', 'SBAS', and 'IRNSS' are unchecked. The 'Station Position File' section has 'Load', 'Save', 'OK', and 'Cancel' buttons.

Satellite and ephemeris settings for HAS alone strategy:



Satellite and ephemeris settings for BHC strategy:

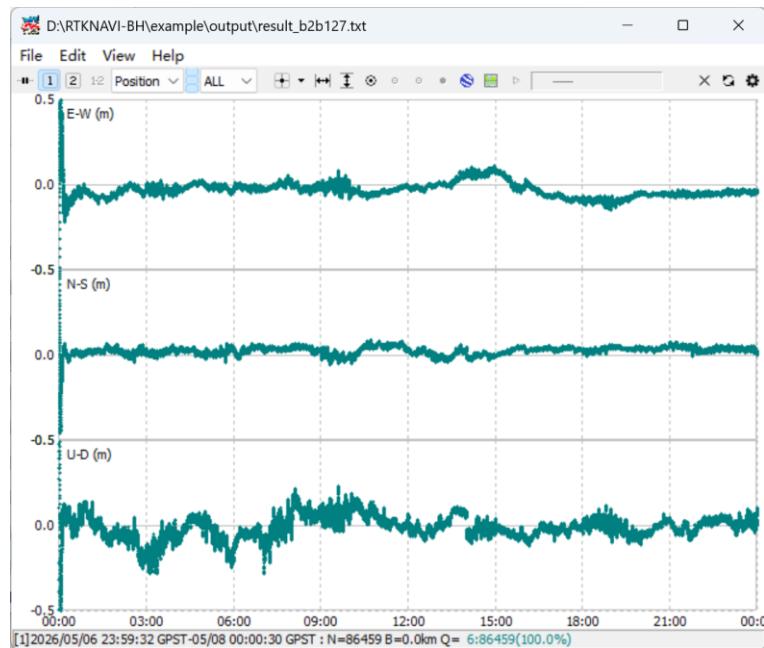


5.6 Starting the processing and viewing results

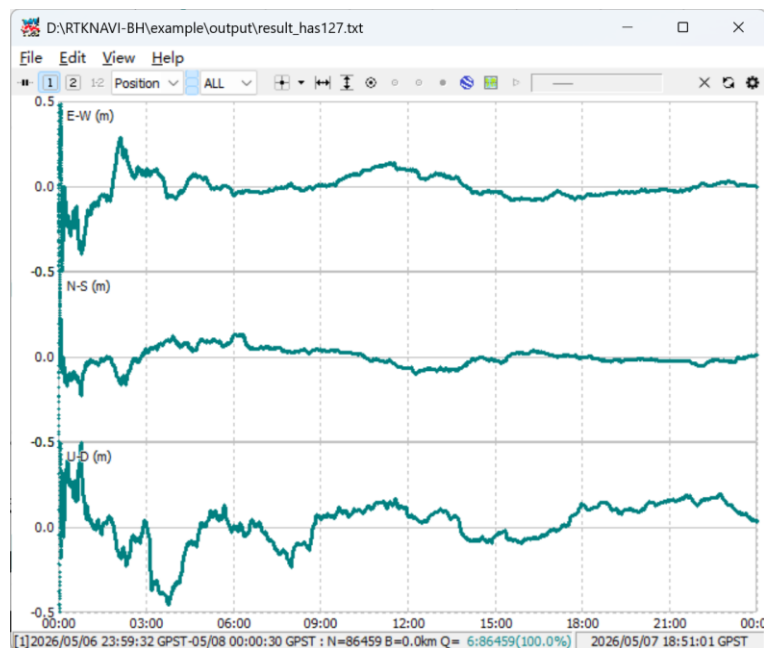
Once all configurations are complete, click the “Start” button on the main window to begin real-time PPP processing. The output files of RTKNAVI-BH are fully compatible with other RTKLIB software. Users can open the positioning results directly in RTKPLOT for visualization, such as plotting error time series and analyzing the convergence process. The real-time PPP results obtained

with the three correction strategies are shown in the figures.

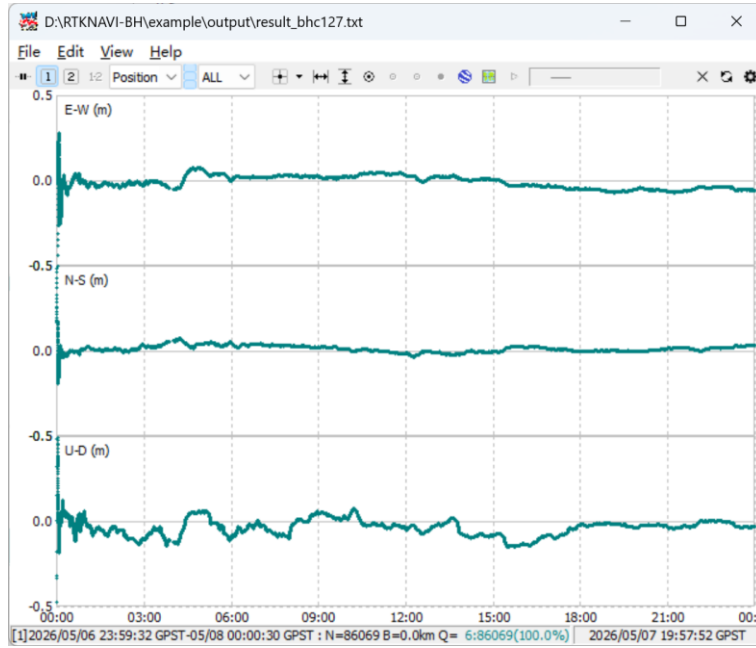
Real-time kinematic PPP positioning error series for PPP-B2b:



Real-time kinematic PPP positioning error series for HAS:



Real-time kinematic PPP positioning error series for BHC:



Appendix

A. Receiver configuration and its raw PPP-B2b and HAS message format

Raw PPP-B2b and HAS messages must be obtained through GNSS receivers and cannot be acquired from IGS/IGMAS data centers. Currently, the extended software supports the raw PPP-B2b and HAS binary messages from SinoGNSS receivers.

A.1. Data output configuration command for SinoGNSS receiver

The following commands configure a SinoGNSS receiver to output the required data streams at 1-second intervals:

unlogall	// Disable all message outputs;
log HasMess ontime 1	// Output HAS messages
log B2BRAWNAVSUBFRAMEb ontime 1	// Output PPP-B2b messages
log gpsephemb ontime 1	// Output GPS broadcast ephemerides
log bd3ephemb ontime 1	// Output BDS-3 broadcast ephemerides
log galephemerisb ontime 1	// Output Galileo broadcast ephemerides
log rangeb ontime 1	// Output raw observations
saveconfig	// Save configuration

A.2. SinoGNSS PPP-B2b and HAS data format

The formats of PPP-B2b and HAS data output by SinoGNSS receivers are presented in Table A.1 and Table A.2, respectively. All fields in each message are arranged in the order of the field names

listed in the corresponding table. The “Message Header” field denotes the header of the data, which is used to identify the data type. The “B2b Message” field contains the original 486-bit binary message of PPP-B2b, while the “HAS Message” field contains the Galileo HAS C/NAV page data of 512 bits, followed by 20 unused trailing bits.

The message header of the SinoGNSS protocol is of fixed length, typically occupying 28 bytes. Its detailed structure is shown in Table 3. The first three bytes, 0xAA, 0x44, and 0x12, serve as the fixed preamble for frame identification. The header length field indicates the length of the header itself, and the message length field gives the total length of the entire message. Both fields facilitate message parsing in the software. The Message ID field identifies the type of the message. For PPP-B2b, the Message ID is 1697, and for HAS, it is 1797. Other data types required for positioning are also distinguished by their respective Message IDs. The Week and Ms fields together form the time stamp of the message. Further details on the SinoGNSS data structure can be found in the SinoGNSS board reference manual.

Table A.1 PPP-B2b Message frame structure output by SinoGNSS receivers (unit: bit)

Message Header	PRN	PRN	Reserved	B2b Message	CRC
224	32	6	6	972	32

Table A.2 HAS Message frame structure output by SinoGNSS receivers (unit: bit)

Message Header	Flag	HAS Page	CRC
224	2	512	32

Table A.3 Structure of binary message header

Field	Field Type	Format	Description	Binary Bytes	Binary Offset
1	Sync	char	Sync byte 0xAA	1	0
2	Sync	char	Sync byte 0x44	1	1
3	Sync	char	Sync byte 0x12	1	2
4	Header Length	uchar	Size of header	1	3
5	Message ID	ushort	Index	2	4
6	Reserved	-		1	6
7	Reserved	-		1	7

8	Message Length	ushort	Length of the message, excluding header and CRC	2	8
9	Reserved	-		2	10
10	Reserved	-		1	12
11	Reserved	-		1	13
12	Week	ushort	GPS week	2	14
13	Ms	uint	Start from the GPS week	4	16
14	Reserved	-		4	20
15	Reserved	ushort	Reserved for internal use	2	24
16	Receiver S/W Version	ushort	Receiver version, valid values from 0 to 65535	2	26

B. PPP-B2b message

To effectively distinguish the contents broadcast in the data field, PPP-B2b messages are categorized into multiple types. The currently defined message types are presented in Table B.1. Message Types 1-4 are the most commonly used, covering satellite mask, satellite orbit corrections and user ranging accuracy index (URAI), code bias corrections, satellite clock corrections. This section briefly introduces these message types, and further detailed specifications can be referred to the official PPP-B2b interface control document (ICD).

Table B.1 PPP-B2b message type definition

Message type (decimal)	Message content
1	Mask of satellite
2	Satellite orbit corrections and user ranging accuracy index
3	Code bias corrections
4	Satellite clock corrections
5	User ranging accuracy index
6	Satellite clock and orbit corrections-combination 1
7	Satellite clock and orbit corrections-combination 2

8-62	Reserved
63	Empty

To ensure the correlation of information content broadcast in different message types, Issue of Data (IOD) is used to identify the data. These IODs include:

- 1) IOD SSR: This field indicates the issue number of State Space Representation (SSR) data, and is broadcast in all currently defined message types. Data across different message types are compatible for combined use if their IOD SSR values are identical; otherwise, the data should not be used in combination. The IOD SSR value is typically updated only upon changes to the system-side configuration.
- 2) IODP: This field denotes the issue number of satellite mask data. It is broadcast in Message Type 1, Message Type 4, Message Type 5 and Message Type 6, and enables users to verify the compatibility of data across the aforementioned message types.
- 3) IODN: This field specifies the issue number of ephemeris and satellite clock parameters broadcast via GNSS downlink signals. It is broadcast in Message Type 2, and allows users to verify the compatibility between the ephemeris and clock parameters from the basic navigation message and the orbit corrections contained in Message Type 2, Message Type 6 and Message Type 7.
- 4) IOD Corr: This field defines the issue number of orbit corrections and clock corrections, and is broadcast in Message Type 2, Message Type 4, Message Type 6 and Message Type 7. For a single satellite, the clock corrections and orbit corrections are compatible for combined use if their corresponding IOD Corr values are identical. In conjunction with the IODN field, users can leverage the IOD Corr field to verify the compatibility between the clock parameters from the basic navigation message and the clock corrections contained in Message Type 4, Message Type 6 and Message Type 7.

(1) Message type 1

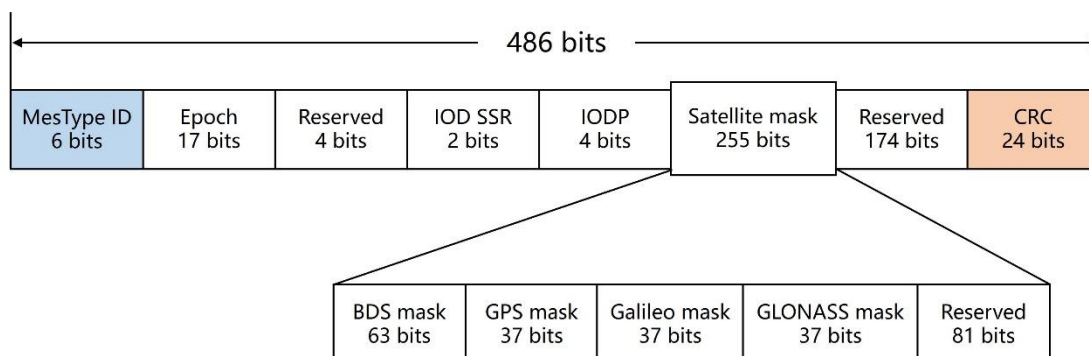


Fig B.1 Layout of PPP-B2b message type 1

Message Type 1 carries satellite mask information. The mask consists of 255 indicator positions, with each position allocated 1 bit. A bit value of 1 indicates that differential correction data for the corresponding satellite is broadcast, while a bit value of 0 indicates that no differential correction data for the satellite is broadcast. The mask blocks corresponding to different GNSSs are arranged in the fixed sequence of BDS, GPS, Galileo, and GLONASS.

(2) Message type 2

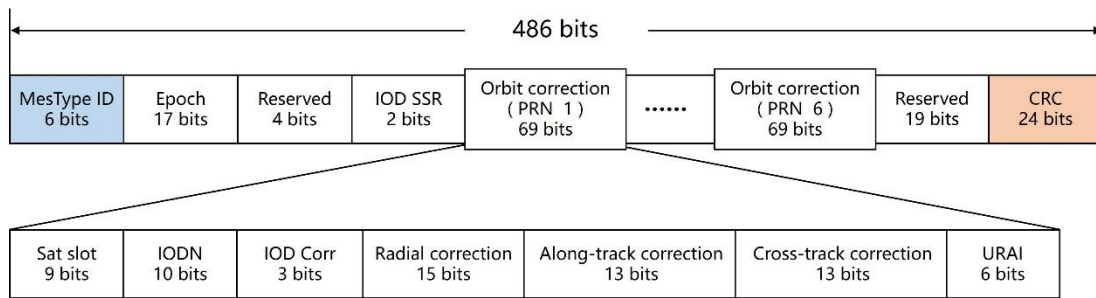


Fig B.2 Layout of PPP-B2b message type 2

Message Type 2 broadcasts orbit corrections for 6 satellites. The orbit correction dataset for each individual satellite comprises the Sat slot, IODN, IOD Corr, as well as radial, along-track and cross-track corrections. The URAI is also broadcast concurrently within this message.

(3) Message type 3

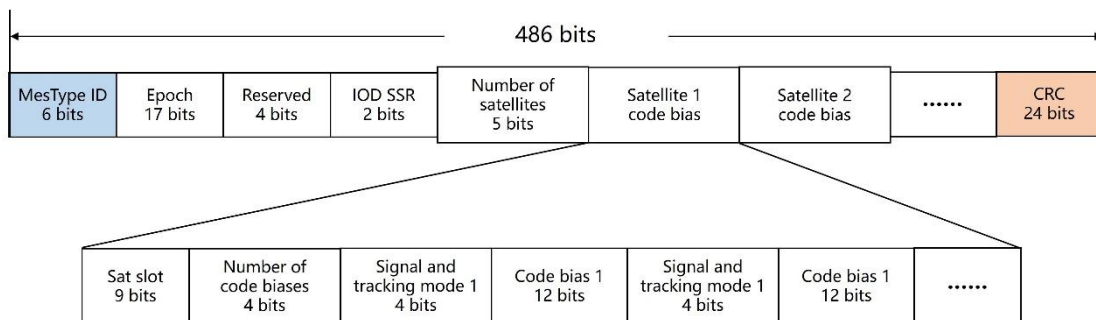


Fig B.3 Layout of PPP-B2b message type 3

Message Type 3 broadcasts code biases for each signal branch of each satellite. The number of satellites included in each message, as well as the number of code biases for each individual satellite, are variable. The Signal and Tracking Mode field identifies the signal branch corresponding to the code bias and the reception mode of the signal on this branch, with its detailed definition specified in the ICD.

(4) Message type 4

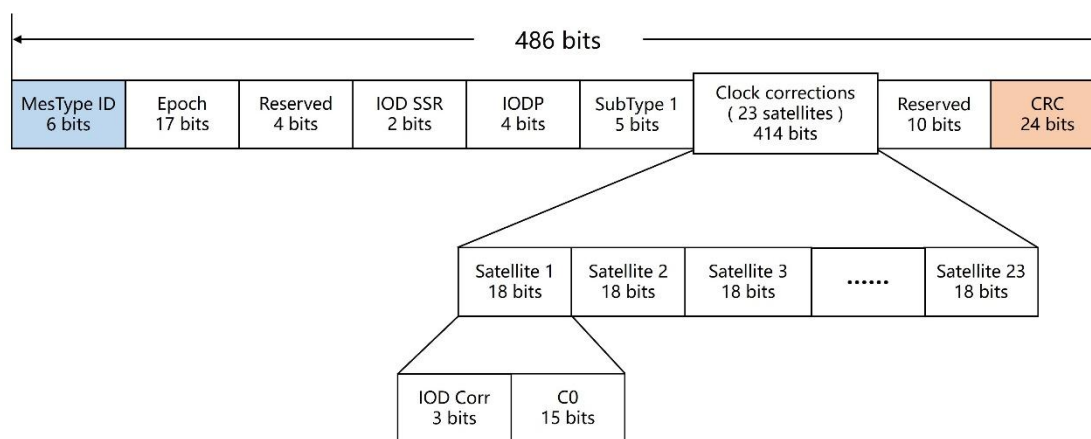


Fig B.4 Layout of PPP-B2b message type 4

For Message Type 4, users jointly identify the satellites corresponding to the clock corrections via the combination of the message type, the mask (broadcast in Message Type 1), and the subtype (SubType1). All satellites with their corresponding mask bit set to 1 should be arranged in a compacted sequence in strict accordance with the order of their positions in the mask.

C. HAS message

C.1. HAS Page

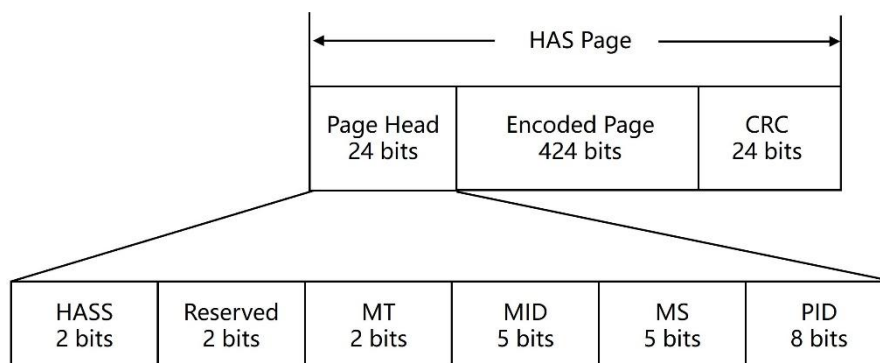


Fig C.1 Layout of HAS page

As shown in Figure C.1, each C/NAV page encapsulates a 448-bit HAS Page. Each HAS Page consists of a 24-bit HAS Page Header and a 424-bit HAS Encoded Page field. HAS Encoded Pages correspond to segmented fragments of encoded HAS Messages. The HAS Page Header comprises the following fields: HASS, MT, MID, MS, and PID. The HASS field specifies the operational status of the HAS service, where: “00” denotes “Test Mode”; “01” denotes “Operational Mode”; “11” denotes “Don’t use”. The Message Type (MT) field currently supports only message type 1, hereafter referred to as MT1.

To reassemble fragmented HAS messages, users should combine the following keywords: MID (ID of message), MS (size of message), and PID (ID of transmitted HAS encoded page). Once MS

plus 1 HAS Encoded Pages with an identical MID are received, the complete MT1 message can be recovered by reassembling and decoding these pages in the sequence defined by their PID values.

C.2. HAS MT1 message

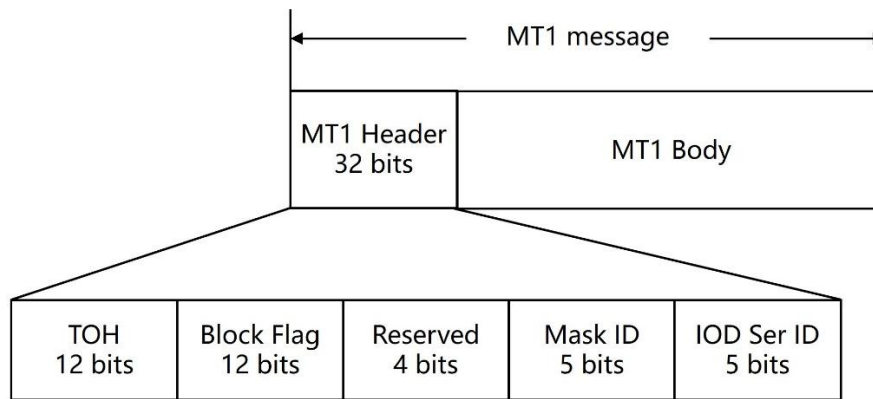


Fig C.2 Layout of HAS MT1 message

The layout of the MT message after HPVRS decoding is shown in Figure C.2. The length of the MT message is variable, with two primary variants formally defined for current operational use: MT1-slow and MT1-fast. The two variants are distinguished via the header field of the MT1 message.

The Time of Hour (TOH) field indicates the time tag of the corresponding message, referenced to Galileo System Time (GST). In combination with the validity interval field contained in each message, the TOH field defines the valid time window of the HAS correction products. The HAS correction products consist of six types of data blocks: Mask, Orbit Corrections, Clock Full-Set Corrections, Clock Subset Corrections, Code Biases, and Phase Biases.

The presence of each data block within the MT message is signaled by the Block Flag field, which indirectly determines the total length of the MT message. For current operational implementations, two core MT1 variants are specified: MT1-slow (the MT1 message carrying Mask, Orbit Corrections, and Code Biases blocks) and MT1-fast (the MT1 message carrying the Clock Full-Set Corrections block).

Complete technical details for all fields and message structures are specified in the HAS ICD. The full set of Galileo HAS satellite orbit, clock correction, and code bias products can be generated by combining MT1-slow and MT1-fast messages through matching their Mask ID and IOD Set ID.

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